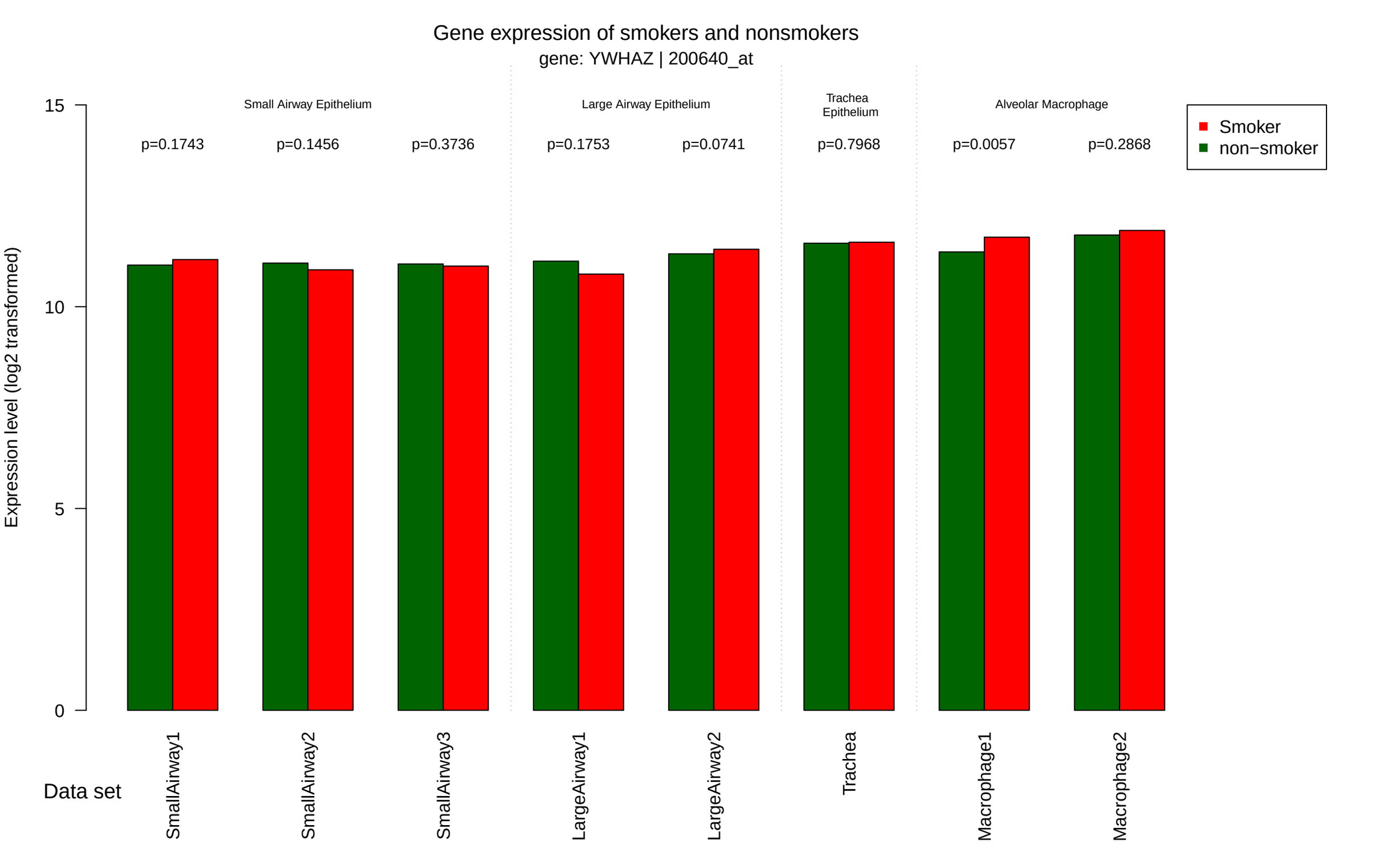
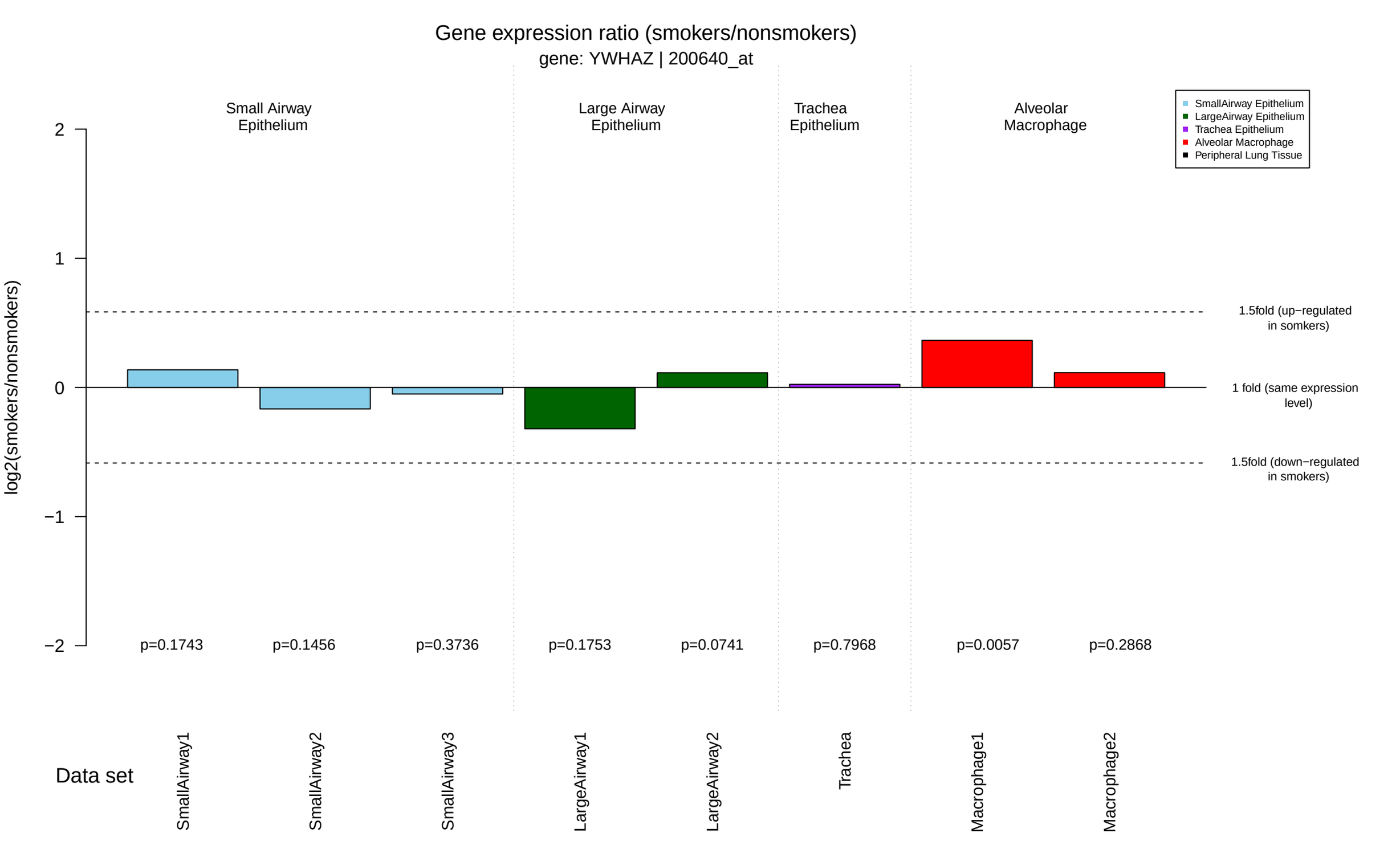
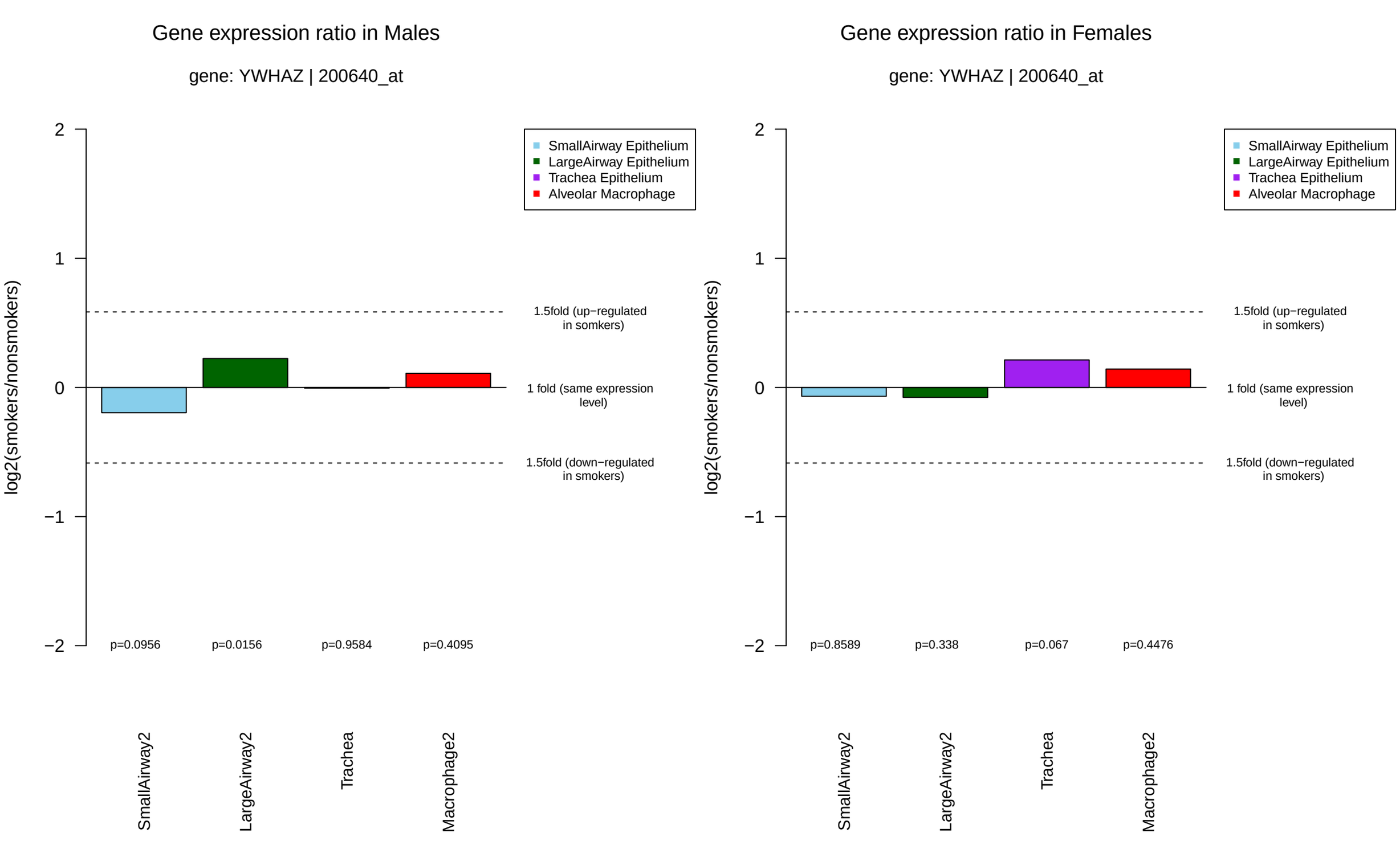
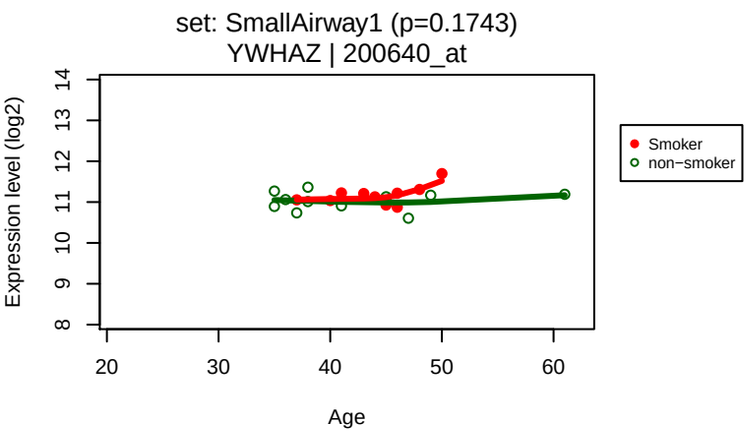
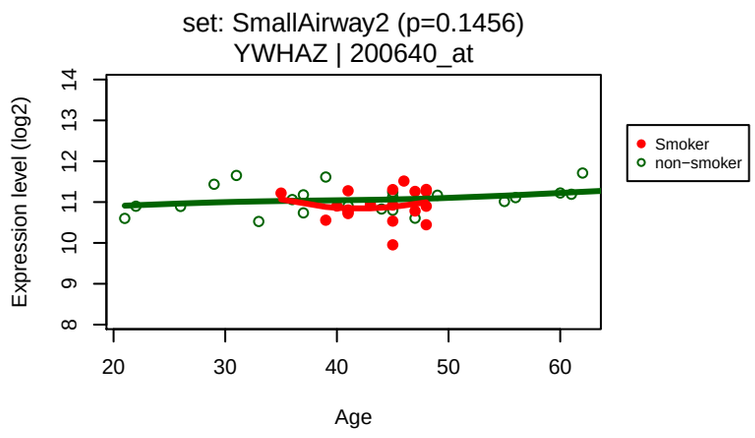
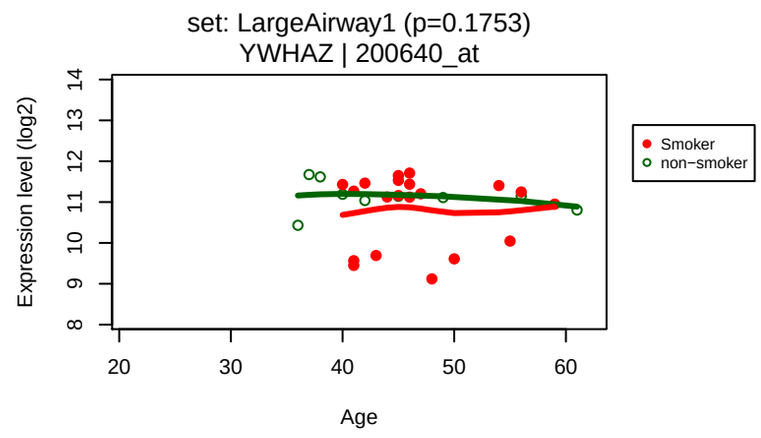

Gene expression of smokers and nonsmokers
gene: YWHAZ | 200640_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1743
p=0.1456
p=0.3736
p=0.1753
p=0.0741
p=0.7968
p=0.0057
p=0.2868
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

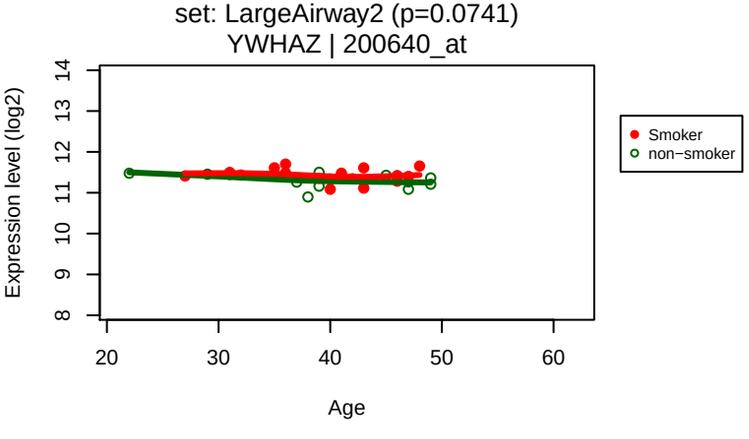

Gene expression ratio (smokers/nonsmokers)
gene: YWHAZ | 200640_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1743
p=0.1456
p=0.3736
p=0.1753
p=0.0741
p=0.7968
p=0.0057
p=0.2868
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

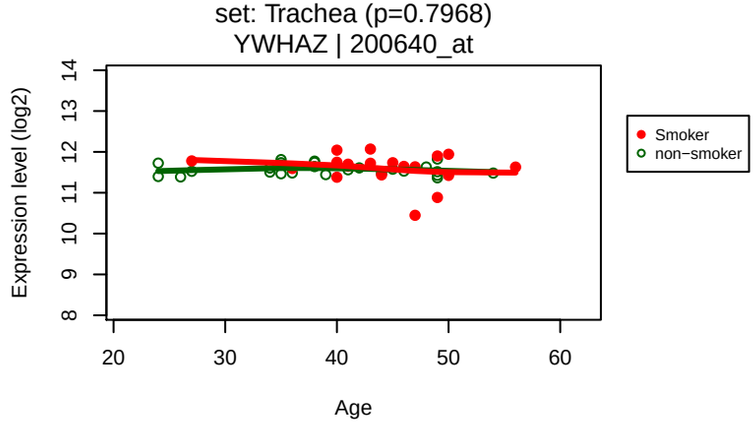

Gene expression ratio in Males
gene: YWHAZ | 200640_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0956
p=0.0156
p=0.9584
p=0.4095
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: YWHAZ | 200640_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8589
p=0.338
p=0.067
p=0.4476
SmallAirway2
LargeAirway2
Trachea
Macrophage2

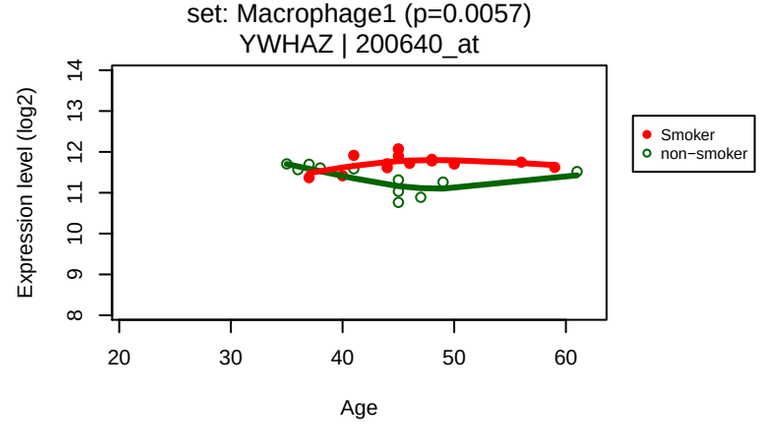

set: SmallAirway1 (p=0.1743)
YWHAZ | 200640_at
Expression level (log2)
Age
Smoker
non−smoker

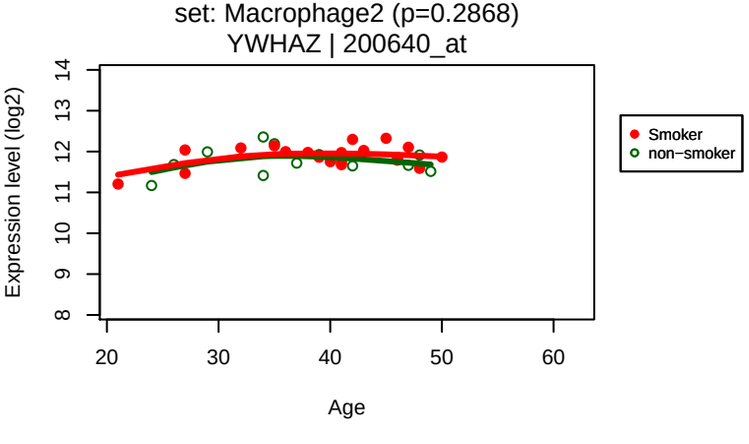

set: SmallAirway2 (p=0.1456)
YWHAZ | 200640_at
Expression level (log2)
Age
Smoker
non−smoker

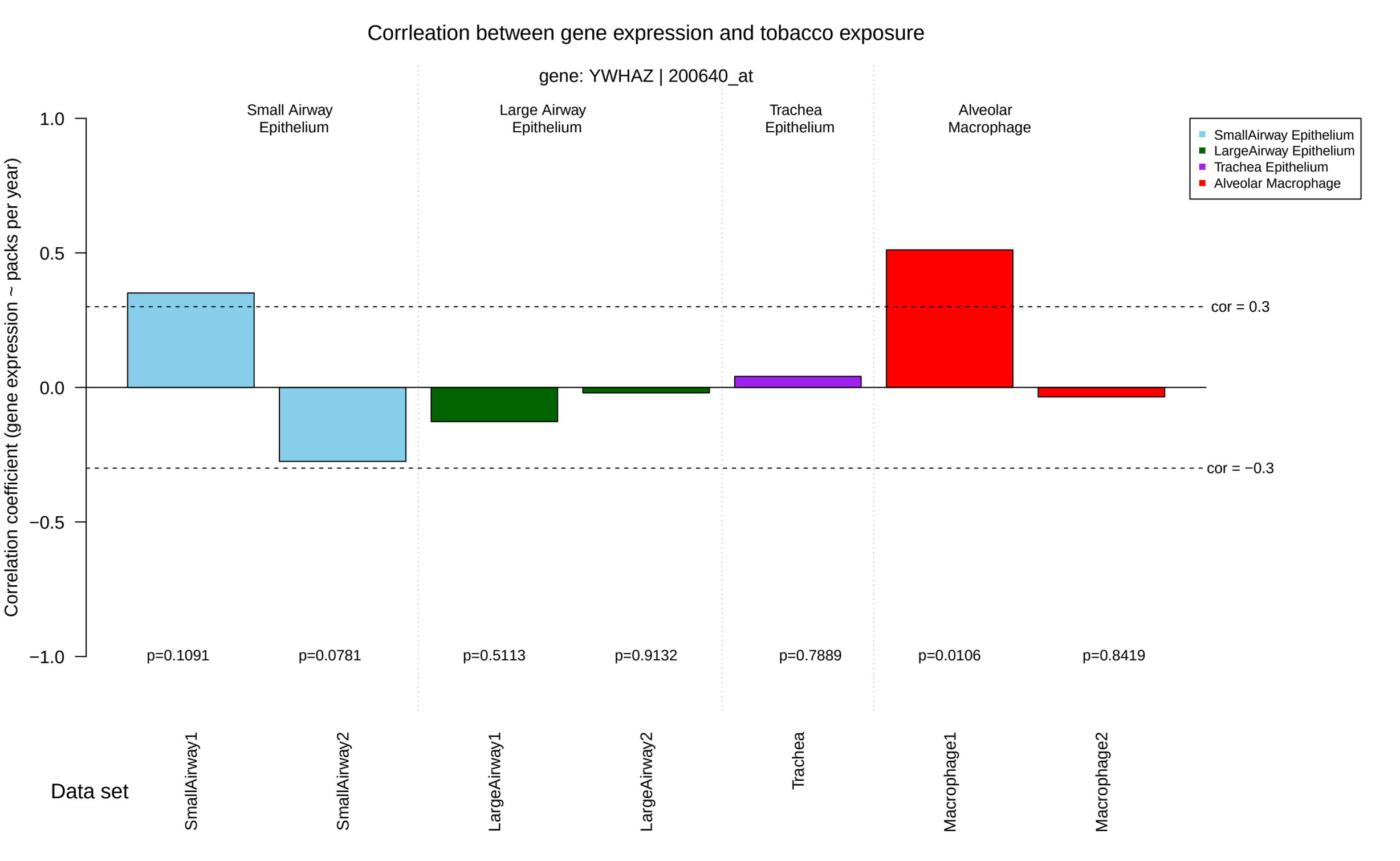

set: LargeAirway1 (p=0.1753)
YWHAZ | 200640_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0741)
YWHAZ | 200640_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7968)
YWHAZ | 200640_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0057)
YWHAZ | 200640_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2868)
YWHAZ | 200640_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: YWHAZ | 200640_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1091
p=0.0781
p=0.5113
p=0.9132
p=0.7889
p=0.0106
p=0.8419
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2